

Projection onto box constraints: Real Stability

Research Architecture Runtime
operator/projection-box

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Abstract

`projection-box` uses interval clamp, which is 1-Lipschitz; hence $|x' - x| \leq \varepsilon$ implies the clamped values move by at most ε . Lean `LEAN_FULLL`.

This theorem is: Reduction.

1 Problem

`projection-box` projects coordinates onto intervals via $\text{clamp}(x; \text{lo}, \text{hi}) = \max(\text{lo}, \min(x, \text{hi}))$.

2 Stability notion

Nonexpansiveness of clamp: output perturbation $\leq$ input perturbation in absolute value.

3 Definitions

Require $\text{lo} \leq \text{hi}$. Stability form is the Lipschitz claim with $|x' - x| \leq \varepsilon$.

4 Theorem

Theorem 1. *If $\text{lo} \leq \text{hi}$ and $|x' - x| \leq \varepsilon$, then $|\text{clamp}(x') - \text{clamp}(x)| \leq \varepsilon$. The Lipschitz constant 1 is sharp.*

5 Intuition

Clipping cannot expand distances: points move weakly toward the interval.

6 Examples

Example 2. $\text{lo} = 0$, $\text{hi} = 1$, $x = 0.4$, $x' = 0.7$, $\varepsilon = 0.3$: both clamp to themselves and move by 0.3.

7 Proof

The map $x \mapsto \max(\text{lo}, \min(x, \text{hi}))$ is 1-Lipschitz on $\mathbb{R}$ when $\text{lo} \leq \text{hi}$ (standard case analysis on the three regions $(-\infty, \text{lo}]$, $[\text{lo}, \text{hi}]$, $[\text{hi}, \infty)$). Therefore $|\text{clamp}(x') - \text{clamp}(x)| \leq |x' - x| \leq \varepsilon$ (Theorem 1). Sharpness: take an interior segment of length ε inside $[\text{lo}, \text{hi}]$.

8 Formal statement

Lean module: `Research.Operators.ProjectionBox.Preservation`. Source file: `lean/Research/Operators/Pro`
Theorems: `projection_box_clamp_stability`, `projection_box_clamp_sharpness`. Uses `Research.Operators.P`
Certificate: `lean/certificates/projection-box/projection-box-clamp-stability`. Status:
LEAN_FULL on Mathlib $\mathbb{R}$ (REAL_MATHLIB).

9 Proof dependencies

- `Projection.Clamp` nonexpansiveness.
- Case analysis on regions.

10 Consequences

Interval/box/clipping operators reduce to clamp. Related: `projection-interval`, `coordinate-clipping`.